

CaptionAI: AI-Powered Social Media Caption Generator

Project Description:

A Comprehensive Measure of Well-Being is a Machine Learning-based application that predicts the Human Development Index (HDI) category of a country using key socio-economic indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per capita. The application classifies countries into one of four HDI categories: Very High, High, Medium, or Low.

Built using Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, and Flask, the system preprocesses the HDI dataset, trains a machine learning model, and provides real-time predictions through a user-friendly web interface. The application helps governments, researchers, policymakers, and development organizations analyze human development efficiently and supports evidence-based planning and policy decisions.

Scenarios:

Scenario 1: Predicting Very High Human Development

A policymaker enters values representing a country with high life expectancy, excellent educational attainment, and high GNI per capita. The system predicts Very High HDI, helping decision-makers understand how strong performance across all development indicators contributes to human well-being.

Scenario 2: Identifying Development Gaps

A government official enters data for a developing country with moderate life expectancy, average years of schooling, and medium GNI per capita. The application predicts Medium HDI and assists policymakers in identifying sectors requiring improvement, such as healthcare, education, or income generation.

Scenario 3: Supporting Development Planning

A researcher analyzes a country with low life expectancy, limited educational opportunities, and low GNI per capita. The system predicts Low HDI, enabling governments and development organizations to prioritize investments and design effective development strategies.

Technical Architecture

Description

The HDI Prediction System follows a three-layer architecture to ensure efficient processing and maintainability.

- **Presentation Layer: A responsive web interface developed using HTML, CSS, and Flask Templates allows users to input HDI indicators and view prediction results.**
- **Application Layer: The Flask backend validates user inputs, preprocesses data, loads the trained machine learning model, and performs HDI prediction using Scikit-learn.**
- **Data Layer: The application uses an HDI CSV dataset for model training and stores the trained model using Joblib for future predictions.**

Pre-requisites

- **Python 3.x**
- **Flask Framework**
- **Scikit-learn**
- **Pandas**
- **NumPy**
- **Matplotlib**
- **Seaborn**
- **Joblib**

- **Jupyter Notebook or Spyder IDE**
- **Anaconda Navigator**
- **Git & GitHub (optional)**

Project Workflow

Milestone 1: Environment Setup

- **Install Python and Anaconda.**
 - **Install Flask, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, and Joblib.**
 - **Configure the development environment.**
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Milestone 2: Dataset Preparation

- **Download the HDI dataset.**
 - **Understand dataset attributes.**
 - **Load the dataset into Pandas.**
 - **Explore the dataset and identify missing values.**
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Milestone 3: Data Preprocessing

- **Remove missing values.**
 - **Encode categorical labels.**
 - **Normalize or preprocess the dataset.**
 - **Prepare training and testing datasets.**
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Milestone 4: Machine Learning Model Development

- **Split the dataset into training and testing sets.**

- **Train the machine learning model using Scikit-learn.**
 - **Evaluate model accuracy.**
 - **Save the trained model using Joblib.**
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Milestone 5: Flask Web Application

- **Create the Flask application.**
 - **Design the HTML interface.**
 - **Accept user input for HDI indicators.**
 - **Predict and display the HDI category.**
 - **Test the application.**
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Milestone 6: Deployment

- **Configure the Flask application.**
 - **Install project dependencies.**
 - **Run the application locally.**
 - **Deploy on Render or PythonAnywhere (optional).**
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Exploring the Website

Home Page

The home page allows users to enter:

- **Life Expectancy**
- **Mean Years of Schooling**
- **Expected Years of Schooling**
- **Gross National Income (GNI) per Capita**

A Predict button submits the information for analysis.

Prediction Result Page

The system displays:

- Predicted HDI Category
- Input Summary
- Model Prediction Result

The prediction can be:

- Very High
- High
- Medium
- Low

Conclusion

A Comprehensive Measure of Well-Being is a machine learning-based application that simplifies Human Development Index prediction using key socio-economic indicators. By integrating Python, Scikit-learn, Flask, Pandas, and NumPy, the system provides accurate and fast predictions that support policymakers, researchers, and development organizations in evaluating human development and making informed decisions. Future enhancements may include integration with live UNDP or World Bank datasets, interactive dashboards, advanced analytics, and cloud deployment to improve accessibility.

